Who Pays the Bill? Administrative Growth and Payroll Adjustment to Revenue Shocks in Chilean Higher Education

Abstract

The argument that colleges and universities suffer from administrative bloat carries two predictions: administration grows faster than the academic enterprise over time, and it is protected when institutions must cut. Research has concentrated on the first prediction. I test both using the Chilean higher education system, which combines high dependence on tuition, institutions with different funding rules, and two system-wide revenue shocks in 2020 and 2022. I draw on a panel of 154 institutions observed between 2017 and 2024, built from public financial summaries that I reconciled record by record against the original releases, and on audited financial statements from four contrasting universities. Descriptively, the management share of the wage bill rose in professional institutes and some private sectors but not in state universities. In first-difference models with year fixed effects, a 10% decline in real revenue is associated with a decline of 7.7 log points in the real wage bill, while a 10% increase is associated with an increase of 1.4 points. The asymmetry is statistically significant, holds when the high-inflation year of 2022 is excluded, and is not anticipated by future revenue changes. During declines, the management share of the wage bill falls rather than being protected; only in state universities does management also gain share when revenue grows. The case evidence suggests that fee-based academic work is the first margin of adjustment. The findings separate two questions that the bloat debate tends to merge.

Keywords: Administrative bloat; Higher education finance; Revenue shocks; Contingent faculty; Cost asymmetry; Chile

1 Introduction

Few claims about higher education are repeated as often as the claim that administration has grown at the expense of teaching. Since the 1990s, studies of American colleges and universities have documented that administrative spending and staffing increased faster than instructional spending (Leslie & Rhoades, 1995; Gumport & Pusser, 1995), and critics have described the result as an "all-administrative university" in which professional managers displace faculty (Ginsberg, 2011). Others have questioned the size of the trend (Hedrick et al., 2009) or attributed rising costs mainly to the labor intensity of higher education rather than to administrative excess (Archibald & Feldman, 2011).

This debate has focused on long-run growth, but the bloat argument also carries a prediction about what happens when money runs out. If administration expands because it is shielded from the pressures that discipline other parts of the institution, it should also be spared when institutions must cut. The burden of retrenchment would then fall on the academic enterprise. This corollary is central to the policy concern behind the bloat debate, since it implies that teaching pays for administration in bad years as well as good ones. It has rarely been tested, because studies of administrative growth seldom observe how budgets respond to revenue shocks.

I separate the two questions. The first concerns long-run allocation: does the management share of the wage bill grow at the expense of the academic share? The second concerns the incidence of adjustment: when revenue falls, who absorbs the cut? The questions are distinct. An institution can expand its administration over a decade and still reduce administrative staff when a shock arrives. Treating them as one question, as the debate often does, makes it difficult to say what the evidence shows.

Chile provides a useful setting for both questions. Its higher education system depends heavily on tuition and household spending (OECD, 2025). It includes state universities, private universities that are members of the Council of Rectors (CRUCH), other private universities, professional institutes, and technical training centers, which operate under dif-

ferent funding rules and labor arrangements. It experienced two system-wide revenue shocks in quick succession: a loss of revenue in 2020, when the social unrest of late 2019 and the COVID-19 pandemic disrupted admissions, enrollment, and tuition payments, and an erosion of real revenue in 2022, when inflation exceeded 11%. And since 2018 all institutions must submit audited financial statements to a regulator, which publishes annual summaries that divide the wage bill into academic, executive, administrative, and other compensation.

I combine a panel of 154 institutions with an analysis of the audited statements of four universities that differ in ownership and funding, and I make three contributions to research on higher education. First, I show that the wage bill is not rigid downward. It falls almost as much as revenue during declines and recovers slowly during expansions, an asymmetry opposite to the one documented for firms in the management accounting literature (Anderson et al., 2003). Second, I show that management is not a protected core during downturns, while evidence of long-run administrative growth is selective and concentrated in particular sectors. Third, the case evidence suggests that the first margin of adjustment is fee-based academic work, which the regulator's categories do not separate, linking the bloat debate to research on the contingent academic workforce (Kezar et al., 2019).

2 Background

2.1 Administrative growth in higher education

The modern debate on administrative growth draws on an older observation about how colleges spend money. Bowen (1980) argued that institutions pursue excellence, prestige, and influence without a natural limit, raise all the money they can, and spend all they raise. In this "revenue theory of costs," spending follows revenue, and the distribution of new resources among activities reflects internal priorities and pressures rather than a standard of efficient production.

Building on this tradition, Leslie & Rhoades (1995) reviewed the evidence available in the

mid-1990s and found that administrative costs had risen faster than instructional costs across American institutions. They considered several explanations, including external regulation, the growth of student services, and the ability of administrators to capture discretionary resources. Gumport & Pusser (1995) documented administrative growth in a large public university system and described it as "bureaucratic accretion," a process in which administrative structures accumulate over time. Rhoades (1998) and Slaughter & Rhoades (2004) placed the growth of non-faculty professionals within the broader restructuring of academic labor and the turn toward market-oriented activities. Ginsberg (2011) gave the argument its most polemical form.

The empirical picture is less settled than the rhetoric. Some studies have questioned how fast administrative costs have actually grown (Hedrick et al., 2009). Archibald & Feldman (2008, 2011) argued that most of the rise in college costs reflects the cost disease described by Baumol & Bowen (1966): teaching depends on the time of highly educated workers, whose wages rise with productivity elsewhere in the economy, so the relative cost of instruction rises even when nothing is wasted. Non-instructional spending is not necessarily unproductive either. Webber & Ehrenberg (2010) found that spending on student services is associated with higher graduation and persistence rates, particularly at institutions with less selective admissions.

Two features of this literature matter here. First, it is concerned almost entirely with long-run trends. Second, it rests largely on American data, in which the boundaries between administrative and academic spending are drawn by national reporting systems. Recent international evidence suggests that the composition of the wage bill varies widely across systems and that part of this variation reflects how staff are classified (OECD, 2025). In Chile, data from the regulator show that the share of administrative staff in the wage bill rose from 31.9% to 36.7% between 2012 and 2024 (SES, 2025).

2.2 Retrenchment and the protected core

The bloat argument implies a second, less examined prediction. If administrative units grow because they are insulated from the pressures that discipline the academic enterprise, the same insulation should protect them during retrenchment. I refer to this as the protected-core corollary. It predicts that, when revenue falls, the management share of the wage bill should remain stable or rise while academic compensation absorbs the cut.

Two bodies of work suggest reasons to doubt this prediction. The first concerns how costs respond to changes in activity. In firms, selling, general, and administrative costs are sticky: they rise more when sales grow than they fall when sales decline by the same amount, because managers retain resources that would be costly to shed and later rebuild (Anderson et al., 2003). Subsequent work has shown that the direction and size of this asymmetry depend on adjustment costs and managerial expectations, and that costs can also fall faster than they rise (Banker & Byzalov, 2014). Stickiness and rigidity are different properties. Rigid costs are slow to adjust in either direction; sticky costs respond less to declines than to increases. A wage bill can be flexible downward and still be slow to rebuild.

The second concerns the academic workforce. Over recent decades, a growing share of teaching in many systems has been delivered by contingent faculty employed per course or on short contracts (Kezar et al., 2019). Contingent appointments lower the cost of adjustment: they can end at the close of a term without severance, and the courses they cover can be consolidated. The expansion of such arrangements is often linked to public management reforms that favored flexible contracts and the transfer of risk from organizations to workers (Hood, 1991). It also has consequences for students. Ehrenberg & Zhang (2005) found that greater reliance on part-time and non-tenure-track faculty is associated with lower graduation rates, and Deming & Walters (2017) found that cuts in public spending on colleges reduce degree completion. If institutions absorb shocks by cutting contingent teaching, the cost of retrenchment falls on instruction even when administration is also cut. Resource dependence theory adds that organizations facing volatile revenue have stronger incentives to keep a large

part of their costs adjustable (Pfeffer & Salancik, 1978), which suggests that the use of this margin should vary with the source of funding.

2.3 The Chilean higher education system

Readers familiar with American or European systems may find the Chilean system unusual in three respects: the weight of private provision, the role of tuition, and the size of the vocational sector. The current structure dates from a 1981 reform that opened higher education to private providers and created three tiers of institutions. Universities may grant degrees and must be organized as non-profit entities. Professional institutes and technical training centers offer professional and technical programs of shorter duration and were not subject to the non-profit requirement (Valenzuela Nieto, 2015). Universities are further divided into state universities, the private universities that belong to the Council of Rectors (CRUCH), most of them founded before 1981 and receiving direct public transfers, and private universities created after 1981. In 2024, the institutions in the panel enrolled about 1.28 million students. Universities accounted for 55% of enrollment (state universities 15%, private CRUCH universities 16%, other private universities 25%), professional institutes for 33%, and technical training centers for 11%. About five of every six students attended a private institution. Chile is among the OECD countries where independent private institutions enroll the largest share of students, and the system is oriented toward teaching: research accounts for about 6% of institutional expenditure, one of the lowest shares in the OECD (OECD, 2025).

Tuition is the main source of revenue. Household spending accounts for half or more of the total spending of public and government-dependent institutions, a legacy of the 1980s reforms (OECD, 2025). Since 2016, a tuition-free policy (*gratuidad*) covers students from lower-income households who enroll in participating institutions, which must meet accreditation and governance requirements. For these students the state pays a regulated tuition set by the government for groups of programs and adjusted each year for inflation, which has covered

between 60% and 90% of list tuition (Williamson et al., 2024). Students outside the policy pay list tuition, often financed with state-guaranteed loans. Accreditation by the National Accreditation Commission conditions access to these funds. Revenue therefore depends on enrollment, on the share of students who pay or are covered by public funds, and on the real value of tuition.

One feature of academic employment matters for this study. Academic staff are hired either on employment contracts, full- or part-time, or on fee-based contracts (*honorarios*). Fee-based contracts are civil service agreements paid against an invoice; they carry no right to severance and typically cover a single course or term. Staff compensation accounted for 59.3% of total institutional spending in 2024 (SES, 2025).

Law 21.091 of 2018 created the Superintendence of Higher Education (SES) and requires every institution, public or private, to submit audited financial statements and a standardized financial reporting form each year (SES, n.d.). Public summaries of these filings were released by the Higher Education Information Service (SIES) of the Ministry of Education for 2016–2019 and by the SES from 2020 onward. State and private institutions follow different accounting standards. The standard for state institutions defines each compensation account; the standard for private institutions uses the same labels but leaves their definition to each institution (SES & CGR, 2022; SES, 2021).

2.4 Why 2020 and 2022

The period studied includes two system-wide revenue shocks of different nature. Because the financial year coincides with the calendar year and the academic year runs from March to December, events in the last months of one year appear in the accounts of the next. The first shock combined two events. The social unrest that began in October 2019 interrupted the end of the academic year and led to a boycott of the national university admission test in January 2020, which delayed the admission process. Two months later, the COVID-19 pandemic closed campuses and moved teaching online. Total higher education enrollment fell

by 3.6% in 2020 and first-year enrollment by 8.2%, the largest annual decline on record; in non-accredited institutions, enrollment fell by 43.6% (El Dinámo, 2020). At the same time, many families stopped paying tuition. In a sample of ten private universities, the share of students with unpaid balances rose by nine percentage points, to 25%, in the first months of the pandemic. Institutions responded with payment deferrals and additional scholarships, and some turned to bank credit lines to finance their monthly cash flow (Pauta, 2020). The Council of Rectors estimated the 2020 cost of the crisis for its 30 universities at CLP 160.9 billion in undergraduate programs alone, including CLP 94.8 billion in lost tuition, about 8% of expected tuition revenue, CLP 18.8 billion in unbudgeted costs of moving teaching online, and CLP 31.9 billion carried over from 2019 in connection with the social unrest (Consejo de Rectores, 2020). The 2020 shock was therefore a loss of revenue in current pesos: in the panel used here, 63% of institutions saw their nominal operating revenue fall that year.

The second shock was different. Enrollment had recovered by 2022 (El Dinámo, 2022), and only 18% of institutions saw their nominal revenue decline. But consumer prices rose by 11.6% on average, after 4.5% in 2021, and tuition, which institutions set in nominal terms before the academic year begins, did not keep pace. Regulated tuition under the tuition-free policy is adjusted annually for inflation, yet the real income that participating universities received from tuition and institutional transfers fell by 8.6% between 2019 and 2022, a decline attributed mainly to inflation (Williamson et al., 2024). The median institution increased its nominal revenue by 7.2%, and 72% of institutions experienced a decline in real revenue. The 2022 shock was thus an erosion of the real value of revenue rather than a loss of cash. This distinction matters for interpreting the results, because a decline in real compensation in 2022 could reflect wages that were not adjusted for inflation rather than deliberate cuts.

Figure 1 shows real operating revenue by type of institution. Aggregate real revenue fell by about 6% in 2020 and by about 2% in 2022, but these system-level figures conceal much larger shocks for individual institutions. Among institutions with real revenue declines, the average decline was 19.4% in 2020 and 11.1% in 2022. These two years are therefore the

natural focus of the analysis, and their different nature is useful: the 2020 shock tests how institutions respond to a sudden loss of cash, and the 2022 shock tests whether the response survives when the loss is mostly a matter of prices.

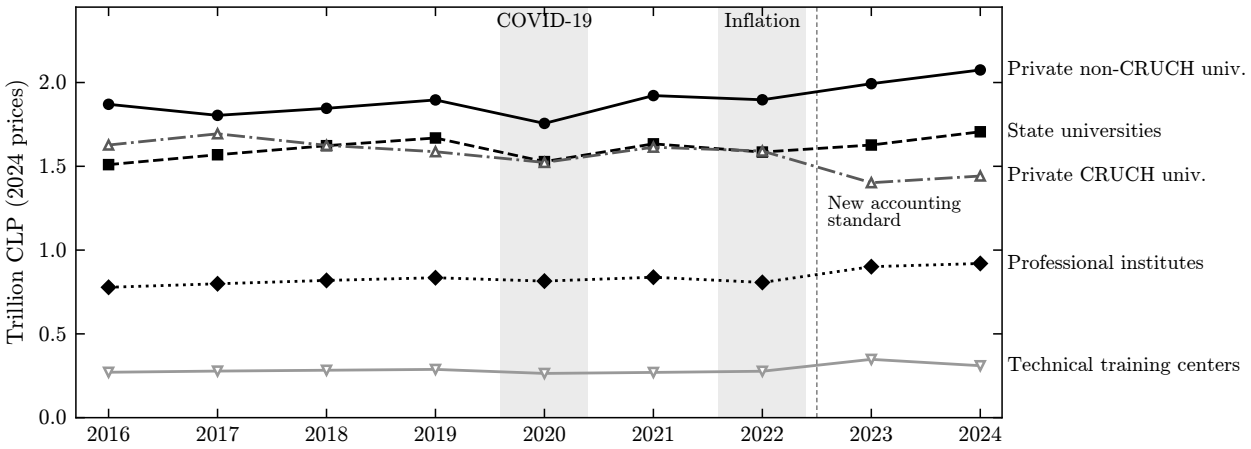


Figure 1: Real operating revenue by type of institution, 2016–2024. Trillion Chilean pesos at 2024 prices, deflated with the annual average consumer price index. The three private universities that joined the CRUCH in 2018–2019 are grouped with private non-CRUCH universities in this figure. Sources: SIES (2016–2019) and SES (2020–2024) financial summaries.

2.5 Research questions

RQ1. Does the management share of the wage bill grow over time at the expense of the academic share?

RQ2. When revenue falls, who absorbs the adjustment? Specifically: (a) does the wage bill respond to revenue declines, and is the response symmetric for increases and declines? (b) Is management protected during declines, as the protected-core corollary predicts? (c) Which contracts are adjusted first?

3 Data

3.1 Panel of institutions

The panel draws on the annual financial summaries published by the SIES (2016–2019) and the SES (2020–2024). Each summary reports operating revenue and compensation in four categories: academic, executive, administrative, and other. I refer to executive and administrative compensation together as management compensation. The panel covers 154 institutions and 1,051 institution-year observations, of which 994 contain a valid breakdown of compensation. Enrollment was obtained from the SIES and accreditation status from the National Accreditation Commission. All monetary values are expressed in 2024 pesos using the annual average consumer price index.

Before estimation, I reconciled every record against the original releases. Two problems required correction.[1] First, the order of the executive and administrative columns changed between releases, so that the two categories are swapped in about nine of every ten records for 2020–2022 when the files are read by position. Executive compensation exceeds administrative compensation in 9 to 19% of records in other years and in about 90% of records in 2020–2022. All analyses use the corrected series. Management compensation, which sums the two categories, is unaffected. Second, a new reporting template introduced with the 2023 financial year changed how institutions classify compensation; the share of other compensation in the wage bill fell from 11.1% in 2022 to 5.4% in 2024. For this reason the main estimation window is 2018–2022, with later years used in robustness checks.

3.2 Audited statements of four universities

The second source is a set of 30 audited financial statements for 2017–2025 from four universities selected after the panel results, following the logic of nested analysis (Lieberman, 2005). The cases vary in ownership and in dependence on public funding (Table 1). The

[1] Revenue and assets matched the original releases in 98 to 100% of cases.

notes to the statements report information that the summaries omit, including compensation by type of contract, severance payments, and in some cases headcount. I reconciled each statement with the regulator's summary for the same institution and year and used press reports to establish the chronology of the most extreme case.

Table 1: Case selection

	Public funding	Tuition dependent
State	Universidad de Antofagasta; Universidad Arturo Prat	–
Private	Universidad Santo Tomás (large, part of an educational group, tuition-free policy since 2022)	Universidad de Aconcagua (no tuition-free policy, not accredited)

4 Methods

4.1 Long-run composition

I address RQ1 descriptively. For each type of institution, I compute the simple average of the management and academic shares of the wage bill by year, restricted to institutions observed in both 2017 and 2024 so that changes in composition are not driven by entry and exit.

4.2 Response to revenue shocks

I address RQ2 with a first-difference model estimated separately for each outcome:

$$\Delta Y_{it} = \alpha + \beta_1 \, Growth_{it} + \beta_2 \, Decline_{it} + \delta \, \Delta \ln Enrollment_{it} + \gamma \, Accredited_{it} + \lambda_t + \varepsilon_{it} \quad (1)$$

The outcome ΔY_{it} is either the change in the real level of a compensation category, measured in log points ($100 \times \Delta \ln$), or the change in the share of that category in the wage

bill, in percentage points. Real revenue growth g_{it} is split into $Growth_{it} = \max(g_{it}, 0)$ and $Decline_{it} = \max(-g_{it}, 0)$, both in units of 10%. The coefficient β_1 is the change in the outcome associated with a 10% increase in revenue, and β_2 the change associated with a 10% decline; a negative β_2 indicates a cut. A symmetric response implies $\beta_1 = -\beta_2$.

The design compares each institution with itself and, through the year fixed effects λ_t, compares institutions that experienced revenue changes of different size in the same year. Shocks common to all institutions in a year, such as the pandemic, national inflation, or regulatory changes, are absorbed by the fixed effects. The change in log enrollment separates financial shocks from changes in activity, so that β_2 captures the response to revenue declines not explained by having fewer students, for example changes in public transfers, tuition, or student aid. Accreditation status conditions access to public funding and student loans. Standard errors are clustered by institution.

Changes in the level of individual compensation categories contain extreme values, often produced when an institution moves staff between categories. In the level models, the outcome is therefore bounded at the 1st and 99th percentiles. Share changes are bounded by construction and are not modified. The main window, 2018–2022, includes 521 observations from 127 institutions with complete data. Level models for individual categories use slightly fewer observations, because categories with zero compensation in a year are excluded.

The model estimates associations rather than causal effects. Two threats deserve particular attention. If institutions cut their wage bills in anticipation of revenue declines, or if cuts themselves reduced revenue, current compensation changes would be correlated with future revenue changes. I test this by adding the next year's revenue change to the model. Because the 2022 shock was mainly an erosion of real values, a decline in real compensation in that year could reflect nominal wages that were not adjusted for inflation rather than active cuts. If so, the asymmetry would be concentrated in 2022. I test this by excluding that year.

4.3 Case analysis

For the four universities, I trace the trajectory of the real wage bill and, where the notes allow, its composition by type of contract. I use the cases to observe mechanisms that the panel cannot detect, in particular the role of fee-based academic contracts, and not to estimate effects.

4.4 Use of generative AI

I used a generative AI tool (Claude, Anthropic) to assist in drafting the manuscript, in writing code to replicate the estimates and produce figures, and in checking references. I designed the study, compiled and reconciled the data, conducted the case analysis, and reviewed and verified all analyses, text, and references.

5 Results

5.1 Long-run composition (RQ1)

Figure 2 and Table 2 show the composition of the wage bill by type of institution. Between 2017 and 2022, the period before the template change, the management share rose in professional institutes (from 46.1% to 49.0%), private CRUCH universities (from 37.3% to 40.7%), and private technical training centers (from 43.1% to 45.0%). It fell in state universities (from 47.5% to 43.7%) and private non-CRUCH universities (from 36.9% to 33.3%). After 2022 it rose in every type of institution, most sharply in professional institutes, where it reached 57.6% in 2024 while the academic share fell from 44.4% in 2017 to 34.4%.

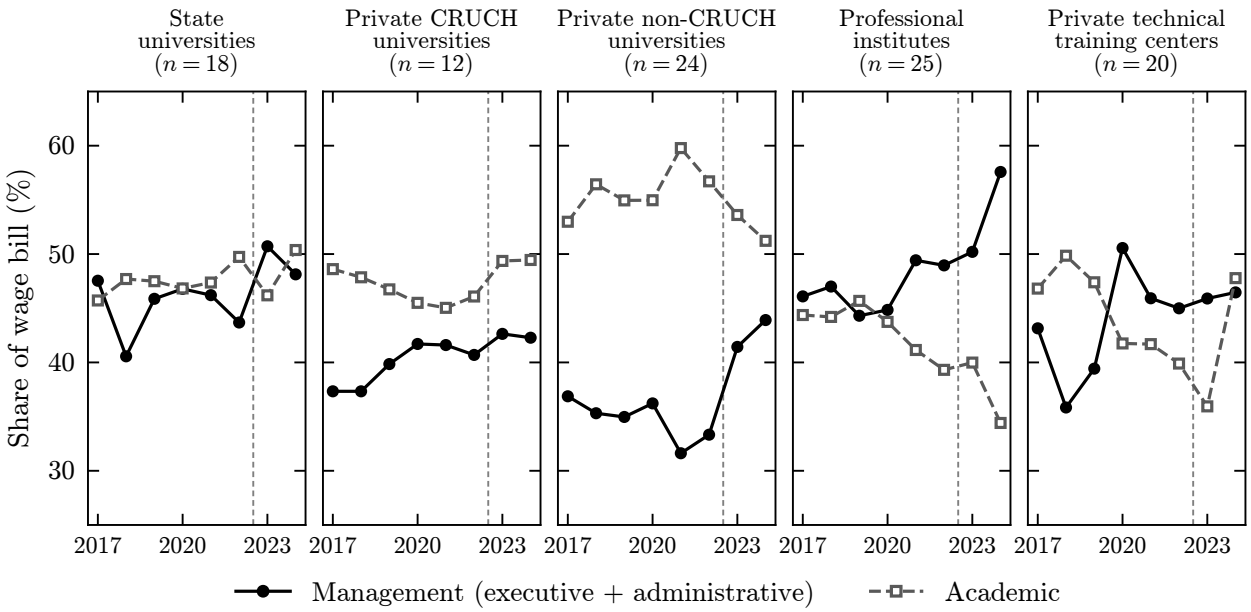


Figure 2: Management and academic shares of the wage bill by type of institution, 2017–2024. Simple averages for institutions observed in both 2017 and 2024. The dotted line marks the reporting template introduced in 2023.

Table 2: Management and academic shares of the wage bill (%), institutions observed in 2017 and 2024

		Management share			Academic share		
Type	n	2017	2022	2024	2017	2022	2024
State universities	18	47.5	43.7	48.1	45.7	49.7	50.4
Private CRUCH universities	12	37.3	40.7	42.3	48.6	46.1	49.5
Private non-CRUCH universities	24	36.9	33.3	43.9	53.0	56.7	51.2
Professional institutes	25	46.1	49.0	57.6	44.4	39.3	34.4
Private technical training centers	20	43.1	45.0	46.5	46.8	39.9	47.8

Note. Simple averages. Management combines executive and administrative compensation. State technical training centers are excluded because they entered the system after 2017. Values for 2024 follow the reporting template introduced in 2023.

The answer to RQ1 is therefore mixed. Administrative growth at the expense of academic compensation is visible in professional institutes and, more weakly, in private CRUCH

universities and private technical centers. It is not visible in state universities or private non-CRUCH universities before 2023. The increases after 2022 coincide with the new template and cannot be separated from reclassification with these data.

5.2 The wage bill falls quickly and recovers slowly (RQ2a)

Table 3 and Figure 3 report the response of real compensation to revenue changes. A 10% decline in real revenue is associated with a decline of 7.74 log points in the real wage bill. A 10% increase is associated with an increase of 1.39 points, which is not statistically different from zero. The hypothesis of a symmetric response is rejected ($p = 0.002$). During declines, academic compensation falls by 5.71 points, management compensation by 6.41 points, and executive compensation by 7.31 points, all statistically significant. The decline in administrative compensation (3.99 points) and in other compensation (4.41 points) is not statistically significant.

Table 3: Response of real compensation to revenue changes (log points per 10% change in real revenue)

	Revenue growth	Revenue decline	Symmetry test (p)	Obs.
Total wage bill	1.39	-7.74^{***}	0.002	521
	(0.88)	(1.83)		
Academic	1.86^{**}	-5.71^{***}	0.112	516
	(0.93)	(2.20)		
Management	4.08^{***}	-6.41^{***}	0.300	493
	(1.28)	(1.90)		
Executive	2.84	-7.31^{***}	0.247	433
	(1.84)	(2.56)		
Administrative	3.85^{**}	-3.99	0.961	489
	(1.96)	(2.44)		
Other	0.71	-4.41	0.511	472
	(3.10)	(4.22)		

Note. Coefficients β_1 and β_2 from Equation 1, window 2018–2022. All models include the change in log enrollment, accreditation status, and year fixed effects. Outcomes bounded at the 1st and 99th percentiles. Standard errors clustered by institution in parentheses. The symmetry test is $\beta_1 = -\beta_2$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

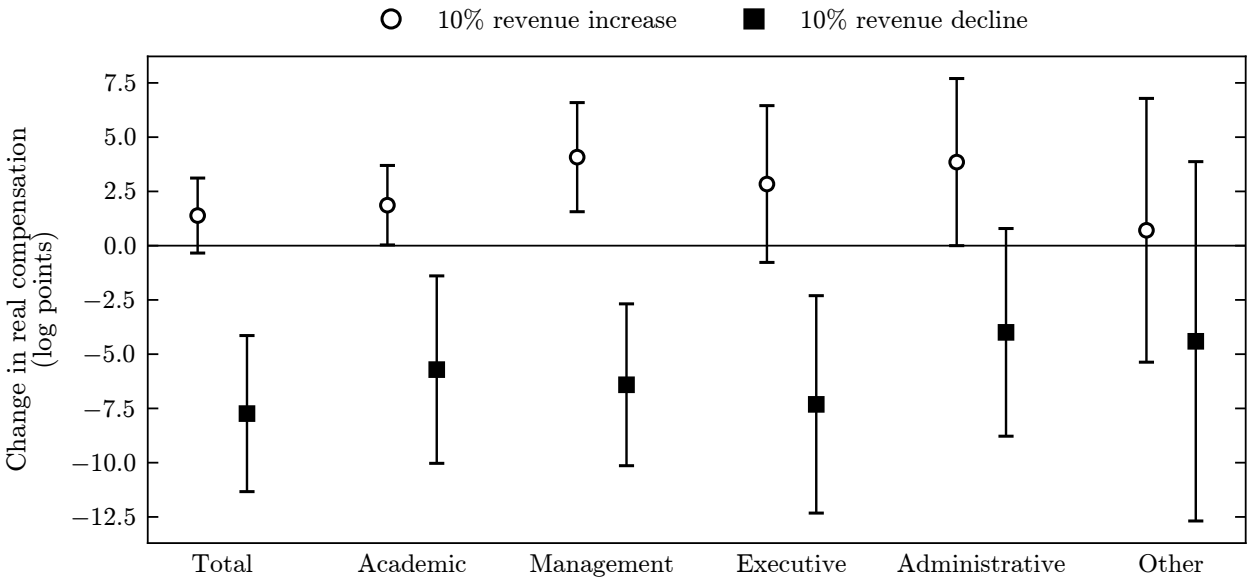


Figure 3: Response of real compensation to a 10% increase and a 10% decline in real revenue, by category. Bars show 95% confidence intervals based on standard errors clustered by institution.

Table 4 shows that the result for the total wage bill is robust. The response to declines ranges from −5.85 to −11.17 log points across specifications and is always statistically significant. Excluding 2022 leaves the response almost unchanged (−7.84), so the asymmetry is not an artifact of nominal wages lagging behind inflation in that year. When the next year's revenue change is added, its coefficient is small and not significant (-0.90 (SE 1.53) for a future decline), while the coefficient on the current decline remains close to the main estimate. Current cuts are not anticipating future revenue changes.

Table 4: Robustness of the response of the total wage bill

Specification	Revenue growth	Revenue decline	Symmetry test (p)	Obs.
(1) Main specification	1.39	-7.74^{***}	0.002	521
	(0.88)	(1.83)		
(2) Not bounded	1.46	-8.85^{***}	0.001	521
	(0.98)	(2.05)		
(3) Percentage change, not bounded	2.75	-8.77^{***}	0.055	521
	(1.91)	(2.49)		
(4) No controls	0.29	-11.17^{***}	0.000	593
	(0.71)	(0.92)		
(5) Excluding 2022	2.30^{**}	-7.84^{***}	0.013	428
	(1.01)	(2.06)		
(6) Window 2018–2024	2.06^{***}	-5.85^{***}	0.010	709
	(0.73)	(1.71)		
(7) With leads of revenue change	1.73^{*}	-7.14^{***}	0.011	509
	(1.03)	(1.79)		

Note. Outcome is the change in the real wage bill in log points unless indicated. Specification (3) uses the percentage change. Specification (4) omits enrollment and accreditation and therefore uses a larger sample. Standard errors clustered by institution in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The wage bill in Chilean higher education is therefore not rigid downward. It falls almost as much as revenue during declines and grows little during expansions. This runs opposite to the sticky pattern documented for administrative costs in firms (Anderson et al., 2003), and it qualifies the revenue theory of costs (Bowen, 1980): institutions in this system do not spend all they raise in good years as quickly as they cut in bad ones.

The result is compatible with the regulator's observation that compensation rose by 0.7% in nominal terms in 2020 while total spending fell (SES, 2025). That figure is nominal and refers to the system as a whole. The model absorbs the common shock and compares institutions that lost different amounts of revenue in the same year, in real terms. A wage

bill that looks stable in the aggregate can coexist with rapid adjustment in the institutions that lose the most.

5.3 Management is not a protected core (RQ2b)

Table 5 reports the response of the shares of the wage bill. In the main specification, a 10% decline in revenue is associated with a fall of 2.69 percentage points in the management share ($p = 0.049$), concentrated in executive compensation. The academic share and the share of other compensation do not change significantly. The management share does not respond to revenue growth (0.09 points, standard error 0.31).

Table 5: Response of wage bill shares to a 10% decline in revenue (percentage points)

Specification	Academic	Executive	Administrative	Management	Other	Obs.
(1) Main specification	0.89	−1.35**	−1.35	−2.69**	1.80	521
	(0.80)	(0.64)	(1.03)	(1.37)	(1.25)	
(2) No controls	−0.32	−0.42**	−0.49	−0.90*	1.23**	593
	(0.33)	(0.21)	(0.45)	(0.48)	(0.51)	
(3) Institutions active in 2024	1.26	−1.52**	−1.79	−3.32**	2.06	484
	(0.86)	(0.75)	(1.18)	(1.50)	(1.59)	
(4) Nominal revenue	0.93	−1.31*	−1.61	−2.92*	1.99	521
	(0.88)	(0.72)	(1.14)	(1.53)	(1.41)	
(5) Outcomes bounded (1st/99th pct.)	0.58	−0.94**	−0.90	−1.90*	0.46	521
	(0.65)	(0.40)	(0.77)	(0.98)	(0.47)	
(6) Excluding 2022	1.03	−1.17	−1.87*	−3.04**	2.01	428
	(0.88)	(0.71)	(1.11)	(1.52)	(1.39)	
(7) Window 2018–2024	0.07	−0.69	−1.53*	−2.22	2.15*	709
	(0.91)	(0.91)	(0.81)	(1.41)	(1.21)	
(8) With lagged revenue change	−0.02	−0.86	1.01	0.15	−0.13	398
	(0.52)	(0.62)	(0.64)	(0.61)	(0.37)	

Note. Coefficient on revenue decline (β_2) from Equation 1, with the change in each share as the outcome. Window 2018–2022 unless indicated. Standard errors clustered by institution in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The negative effect on the management share holds in seven of eight specifications, with point estimates between −0.90 and −3.32. It is not significant in the wider 2018–2024 window, which crosses the template change, and it disappears when the previous year's revenue change is added, a specification that loses about a quarter of the observations. The magnitude depends on the sample. Specification (2), without controls, adds about 70 observations that lack enrollment or accreditation data and yields a smaller coefficient; most of the difference comes from these additional observations rather than from the controls themselves. A placebo test that adds the next year's revenue change yields a coefficient of

0.51 (SE 0.58) for a future decline.

The protected-core corollary is therefore not supported. When revenue falls, management loses rather than gains weight in the wage bill. This is a finding about the incidence of adjustment, not about long-run growth, and it is consistent with the descriptive evidence: the management share can rise over time in some sectors while still falling in the years in which revenue declines.

5.4 Differences across types of institution

Table 6 and Figure 4 report the share models estimated separately by type of institution. The estimates are exploratory, given the small size of some groups and the number of comparisons. In state universities, the management share falls by 5.83 percentage points per 10% decline in revenue ($p < 0.10$) and rises by 6.32 points per 10% increase ($p < 0.05$). Management in these institutions follows revenue in both directions. It expands when revenue grows, as the bloat argument predicts, but it is not protected when revenue falls. In private CRUCH universities, the academic share rises by 3.35 points per 10% decline ($p < 0.01$), the most precisely estimated coefficient in the table. In these universities the composition of the wage bill moves toward academic compensation during downturns, either because academic staff are protected or because other categories are cut first. In the remaining types, no coefficient on revenue decline is statistically significant.

Table 6: Response of wage bill shares by type of institution (percentage points per 10% change in real revenue)

	Academic		Management		Other		
Type	Growth	Decline	Growth	Decline	Growth	Decline	Obs.
State universities	−1.98	−1.66	6.32**	−5.83*	−4.35*	7.49**	86
	(3.04)	(2.13)	(2.67)	(3.06)	(2.34)	(3.65)	
Private CRUCH universities	−0.11	3.35***	0.05	−2.23	0.06	−1.13	58
	(0.55)	(1.10)	(0.61)	(1.50)	(0.82)	(1.35)	
Private non-CRUCH universities	0.39	1.29	−0.52	−1.72	0.13	0.43	120
	(1.18)	(1.38)	(1.13)	(1.70)	(0.44)	(0.67)	
Professional institutes	0.23	2.76	−0.94	−2.13	0.71	−0.63	134
	(0.68)	(2.78)	(0.60)	(2.44)	(0.89)	(0.81)	
Private technical training centers	0.47*	−0.21	−0.07	−1.45	−0.40	1.67	113
	(0.27)	(0.67)	(0.17)	(1.87)	(0.27)	(1.77)	

Note. Window 2018–2022. Coefficients on revenue growth (β_1) and decline (β_2) from Equation 1, estimated separately by type. Standard errors clustered by institution in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

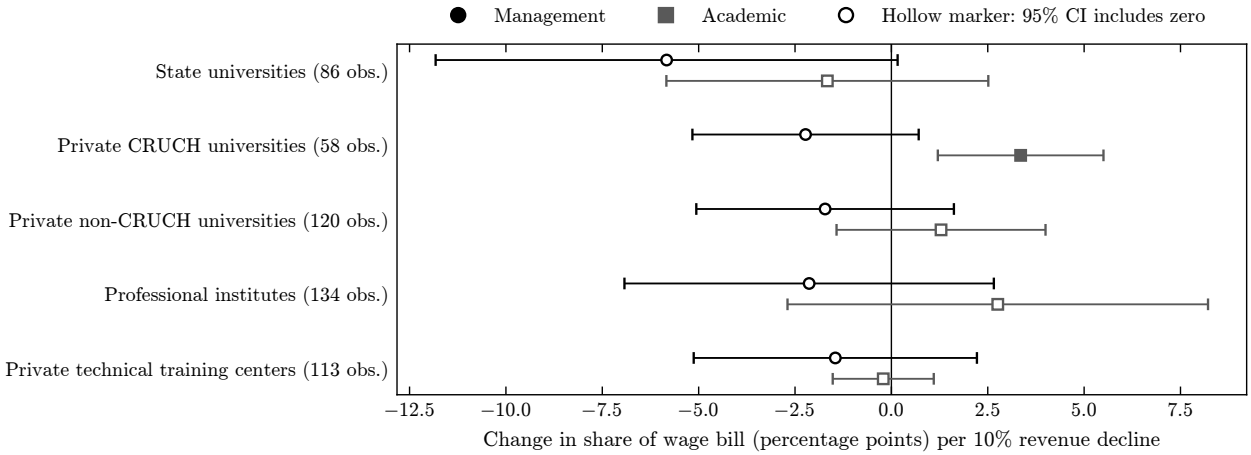


Figure 4: Response of the management and academic shares of the wage bill to a 10% decline in revenue, by type of institution. Bars show 95% confidence intervals; hollow markers indicate intervals that include zero.

The pattern in state universities is the one that comes closest to the bloat argument, and it differs from it in an instructive way. Administrative units in these institutions grow with

resources, as Bowen (1980) and Gumport & Pusser (1995) would expect, but they do not behave as a protected core. The evidence for professional institutes is also instructive. Their management share rose more than in any other sector over the long run (Table 2), yet their management share does not rise when their revenue grows. Long-run administrative growth in this sector does not appear to be driven by revenue windfalls.

5.5 Evidence from four universities (RQ2c)

The panel cannot observe contract types, because fee-based academic work is reported within academic compensation together with salaried academic staff. The audited statements make it possible to look inside this category.

Figure 5 shows the real wage bill of the four universities. In 2020, the Universidad Santo Tomás reduced its wage bill by 13% and rebuilt it the following year as revenue recovered. The Universidad de Antofagasta reduced its wage bill by 10% in 2020 and kept it roughly stable afterward. The Universidad Arturo Prat adjusted with a lag: its wage bill fell by 15% in 2020 and by a further 19% in 2021. The Universidad de Aconcagua cut its wage bill by 46% in 2020. Taken together, the cases suggest that the speed and depth of adjustment vary with the source of funding, with the tuition-dependent private university cutting immediately and deeply and the state universities adjusting later. Four cases cannot establish this pattern, but they are consistent with the expectation from resource dependence theory (Pfeffer & Salancik, 1978).[2]

[2]In several case-years, the revenue change recorded in the public series differs from the change computed from the audited statements, mainly because the statements restate the previous year while the public series keeps the figure originally reported. For this reason the case analysis relies on the audited statements.

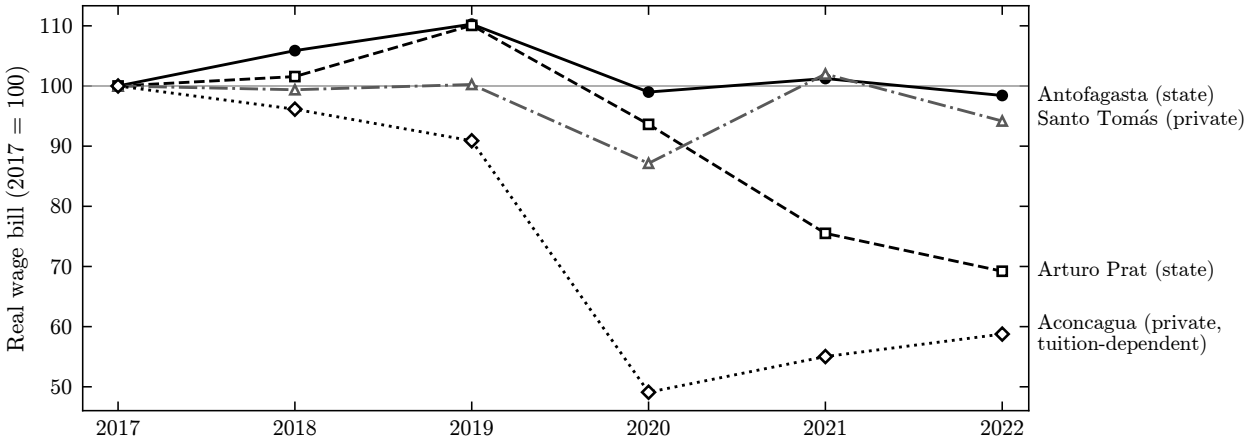


Figure 5: Real wage bill of the four case universities, 2017–2022 (2017 = 100). The series ends in 2022 because the 2023 template changed how some institutions record fee-based teaching.

Where the notes to the statements distinguish types of contract, fee-based academic contracts are reduced first, and salaried staff later or only in deep crises. The 2023 template made fee-based work visible in the notes for the first time and shows how large this margin is (Table 7). Fee-based contracts represented between 15.0% and 33.1% of compensation in the four universities in 2023. The template also shows that institutions classify this work differently. The state universities record fee-based teaching within compensation, the Universidad de Aconcagua records it in a separate line outside compensation, and the Universidad Santo Tomás does both. The regulator's public summary omits the separate line; in the case of the Universidad de Aconcagua, the omitted line amounts to about 30% of the wage bill. Contingent academic work is thus the most flexible part of the academic workforce and also the least visible in the data used to monitor institutions.

Table 7: Fee-based work in the 2023 reporting template

	Antofagasta	Arturo Prat	Santo Tomás	Aconcagua
Composition of the wage bill, 2023 (%)				
Fee-based contracts	15.0	17.6	18.0	33.1
Executives	8.3	6.8	4.4	11.8
Reporting practice				
Fee-based teaching within compensation	Yes	Yes	Partly	No
Headcount reported	No	No	Yes	Yes

The Universidad de Aconcagua shows what adjustment looks like when it reaches salaried staff. Its real wage bill fell by 46% between 2019 and 2020, academic compensation by 43%, and management compensation by 33%. Severance payments in 2020 amounted to CLP 0.57 billion, 14% of the wage bill of that year. Severance is paid to employees on employment contracts, not to fee-based workers, so these payments show that the adjustment reached salaried staff. The crisis began earlier. During 2017 and 2018 the local press reported repeated delays in payments to fee-based faculty, some of whom left, while administrative staff reported delays of only a few days (El Rancagüino, 2018). Enrollment fell by 29% between 2018 and 2020. In March 2020 control of the university passed to another private university, and the press reported 21 dismissals of staff at two campuses the following month (El Aconcagua, 2020). The sequence fits the pattern observed in the other cases: fee-based academic work bore the first adjustment, and salaried staff were cut when revenue, enrollment, and ownership changed at once. Because the dismissals coincided with a change of control, part of the adjustment may reflect restructuring associated with the transfer.

6 Discussion

The two questions that organize this study receive different answers. On long-run allocation, the evidence of administrative growth at the expense of academic compensation is selective.

It is strongest in professional institutes and weaker in some private sectors, and it is absent in state universities and private non-CRUCH universities before the template change. On the incidence of adjustment, the answer is clearer. The wage bill responds quickly to revenue declines and slowly to increases, management is not spared, and the first margin of adjustment appears to be fee-based academic work.

These results suggest that the bloat argument and its protected-core corollary should be evaluated separately. Administration can grow over time without being protected during crises. In Chilean state universities, management expands when revenue grows and contracts when it falls, a pattern closer to Bowen's revenue theory of costs than to the image of an insulated administrative core. In professional institutes, where long-run administrative growth is most pronounced, that growth is not tied to revenue windfalls, which points to other drivers. One plausible candidate is the expansion of online education. Online programs grew from 1.7% of total higher education enrollment in 2015 to 12.4% in 2024 (Parro, 2025), and several of the largest providers of online programs are professional institutes (La Tercera, 2021). Online delivery changes the composition of instructional costs. It allows larger sections and the reuse of course materials, which lowers the cost of instruction per student (Deming et al., 2015), and it unbundles the faculty role, shifting tasks such as course design, platform management, tutoring, and student support to staff who are not classified as faculty (Gehrke & Kezar, 2015). If those staff are reported as administrative, the management share of the wage bill rises without any growth of administration in the sense that the bloat literature intends: what looks like bloat would instead reflect a change in how teaching is produced. The timing is consistent with this explanation. In the professional institutes observed throughout the period, the management share remained between 44% and 47% from 2017 to 2020 and rose to 49.4% in 2021, the year after the pandemic moved teaching online, while the academic share fell from 43.7% to 41.1%. The financial data do not report enrollment by modality, so I cannot test this explanation here. Other drivers are also plausible, including the accreditation and reporting requirements introduced by the 2018

reform, the expansion of student services, and the professionalization of management (Leslie & Rhoades, 1995; Rhoades, 1998), as well as the 2023 template change, which coincides with the largest single increase.

The asymmetry of the wage bill speaks to the literature on costs in higher education. Studies of the cost disease emphasize that teaching is labor intensive and difficult to make more productive (Archibald & Feldman, 2011). The results here suggest that labor intensity does not imply rigidity. In a system where a large share of teaching is delivered under fee-based contracts, the wage bill contains a component that can be reduced quickly and without severance. The flexibility that public management reforms promoted (Hood, 1991) and that studies of the contingent academic workforce document (Kezar et al., 2019) becomes, in this reading, the mechanism through which institutions absorb shocks.

This has implications for students. If the cuts that follow a revenue shock fall first on contingent teaching, the burden of retrenchment falls on instruction even when management is also reduced. Evidence from the United States links greater reliance on contingent faculty to lower completion (Ehrenberg & Zhang, 2005) and links cuts in institutional spending to lower attainment (Deming & Walters, 2017). Whether similar effects follow from the adjustments documented here is an open question.

For those who monitor the financial health of institutions, two implications follow. Indicators based on operating margins can register a recovery that conceals very different forms of adjustment; a margin restored by cutting fee-based teaching is not equivalent to one restored by reducing administrative spending. Monitoring the composition of the wage bill, including fee-based contracts, would make these differences visible. In addition, the data on which monitoring relies require attention: common rules for classifying fee-based work, documentation of changes in reporting templates, and the use of restated figures in public series would make comparisons across institutions and over time more reliable.

This analysis has limitations. The panel model estimates associations. Year fixed effects absorb common shocks, and the placebo and inflation tests address two specific threats, but

the model does not rule out shocks specific to an institution and year, such as a change in leadership or ownership. The magnitude of the effect on the management share depends on the sample. The compensation categories are defined only in the accounting standard for state institutions, so comparisons across sectors compare labels that may not mean the same thing. The descriptive evidence on long-run composition relies on simple averages and is affected by the 2023 template. The evidence on contract types comes from four universities.

Several lines of research follow. Linking the financial data to administrative records of academic staff would make it possible to measure contingent work across the whole system and to test whether it is the main margin of adjustment beyond the four cases. Linking enrollment by modality, which the national information system records for each program, to the composition of the wage bill would test whether the growth of online education explains the rise of management in professional institutes; data on administrative functions would help separate this explanation from others. Heterogeneity by region and by accreditation history deserves attention, although accreditation is itself partly an outcome of financial health. Access to credit is another promising direction. Institutions with access to long-term bank credit can smooth shocks, while those without it may rely on short-term financing or on fixed commitments such as leases, which reduce the share of costs that can be adjusted and may increase the pressure on the wage bill.

7 Conclusion

The claim that administration grows at the expense of teaching has two parts, and they deserve to be tested separately. In Chilean higher education, administrative growth over time is real in some sectors and absent in others. When revenue falls, however, management is not a protected core. The wage bill falls quickly, management shares in the cut, and the first margin of adjustment appears to be fee-based academic work, a category that public financial data do not separate. The question of who pays the bill in bad years has a different

answer from the question of who gains in good ones, and both answers depend on a part of the academic workforce that is easy to cut and hard to see.

Statements and Declarations

Funding, competing interests, and acknowledgments. Provided on the separate title page to preserve anonymity.

Data availability. The financial summaries are published by the Higher Education Information Service of the Ministry of Education (2016–2019) and by the Superintendence of Higher Education (2020–2024). The audited financial statements are publicly available from the Superintendence of Higher Education. The compiled panel, the reconciliation files, and the code used to produce the results are provided as supplementary material.

References

Anderson, M. C., Banker, R. D., & Janakiraman, S. N. (2003). Are selling, general, and administrative costs "sticky"? *Journal of Accounting Research*, *41*(1), 47–63. https://doi.org/10.1111/1475-679X.00095

Archibald, R. B., & Feldman, D. H. (2008). Explaining increases in higher education costs. *The Journal of Higher Education*, *79*(3), 268–295.

Archibald, R. B., & Feldman, D. H. (2011). *Why does college cost so much?* Oxford University Press.

Banker, R. D., & Byzalov, D. (2014). Asymmetric cost behavior. *Journal of Management Accounting Research*, *26*(2), 43–79.

Baumol, W. J., & Bowen, W. G. (1966). *Performing arts: The economic dilemma*. Twentieth Century Fund.

Bowen, H. R. (1980). *The costs of higher education: How much do colleges and universities spend per student and how much should they spend?* Jossey-Bass.

Consejo de Rectores de las Universidades Chilenas. (2020). *Impacto financiero en las universidades del CRUCH por la emergencia sanitaria y social del Covid-19* [Report of the Economic Commission, presented at the session of June 25, 2020].

Deming, D. J., Goldin, C., Katz, L. F., & Yuchtman, N. (2015). Can online learning bend the higher education cost curve? *American Economic Review*, *105*(5), 496–501. https://doi.org/10.1257/aer.p20151024

Deming, D. J., & Walters, C. R. (2017). *The impact of price caps and spending cuts on U.S. postsecondary attainment* (NBER Working Paper No. 23736). National Bureau of Economic Research. https://doi.org/10.3386/w23736

El Dinámo. (2020, June 15). Boicot a la PSU y coronavirus: Las razones que explicarían la baja inédita en matrículas en la educación superior. *El Dinámo*. https://www.eldinamo.cl/educacion/2020/06/15/coronavirus-chile-boicot-a-la-psu-y-coronavirus-las-razones-que-explicarian-la-baja-inedita-en-matriculas-en-la-educacion-superior/

El Dinámo. (2022, July 12). Matrícula de educación superior se recupera y alcanza 1,3 millones de estudiantes. *El Dinámo*. https://www.eldinamo.cl/educacion/2022/07/12/matricula-de-educacion-superior-se-recupera-y-alcanza-13-millones-de-estudiantes/

Ehrenberg, R. G., & Zhang, L. (2005). Do tenured and tenure-track faculty matter? *Journal of Human Resources*, *40*(3), 647–659. https://doi.org/10.3368/jhr.40.3.647

El Aconcagua. (2020, April 1). Universidad de Aconcagua pasó a control de la Universidad Bolivariana. *El Aconcagua*. https://www.elaconcagua.cl/2020/04/01/universidad-de-aconcagua-paso-a-control-de-la-universidad-bolivariana/

El Rancagüino. (2018, July 17). Problemas con los pagos complicarían a la Universidad de Aconcagua. *El Rancagüino*. https://www.elrancaguino.cl/?p=35947

Gehrke, S., & Kezar, A. (2015). Unbundling the faculty role in higher education: Utilizing historical, theoretical, and empirical frameworks to inform future research. In M. B. Paulsen (Ed.), *Higher education: Handbook of theory and research* (Vol. 30, pp. 93–150). Springer.

Ginsberg, B. (2011). *The fall of the faculty: The rise of the all-administrative university and why it matters*. Oxford University Press.

Gumport, P. J., & Pusser, B. (1995). A case of bureaucratic accretion: Context and consequences. *The Journal of Higher Education*, *66*(5), 493–520.

Hedrick, D. W., Wassell, C. S., & Henson, S. E. (2009). Administrative costs in higher education: How fast are they really growing? *Education Economics*, *17*(1), 123–137.

Hood, C. (1991). A public management for all seasons? *Public Administration*, *69*(1), 3–19. https://doi.org/10.1111/j.1467-9299.1991.tb00779.x

Kezar, A., DePaola, T., & Scott, D. T. (2019). *The gig academy: Mapping labor in the neoliberal university*. Johns Hopkins University Press.

La Tercera. (2021). Pandemia consolida el estudio de carreras a distancia: Es el área que más crece en educación superior. *La Tercera*. https://www.latercera.com/que-pasa/noticia/pandemia-consolida-el-estudio-de-carreras-a-distancia-es-el-area-que-mas-crece-en-educacion-superior/ZFCV742PIFHVDABEXELVBMNTDA/

Leslie, L. L., & Rhoades, G. (1995). Rising administrative costs: Seeking explanations. *The Journal of Higher Education*, *66*(2), 187–212.

Lieberman, E. S. (2005). Nested analysis as a mixed-method strategy for comparative research. *American Political Science Review*, *99*(3), 435–452.

OECD. (2025). *The financial sustainability of higher education: Insights from policy in OECD countries*. OECD Publishing. https://doi.org/10.1787/f544ccfe-en

Parro, S. (2025, February 19). Por qué sigue creciendo la educación online. *Pauta*. https://www.pauta.cl/dato-en-pauta/2025/02/19/por-que-sigue-creciendo-la-educacion-online.html

Pauta. (2020, May 14). Cómo la crisis está afectando a las universidades en sus niveles de morosidad. *Pauta*. https://www.pauta.cl/nacional/como-la-crisis-esta-afectando-a-las-universidades-en-sus-niveles-morosidad

Pfeffer, J., & Salancik, G. R. (1978). *The external control of organizations: A resource dependence perspective*. Harper & Row.

Rhoades, G. (1998). *Managed professionals: Unionized faculty and restructuring academic labor*. State University of New York Press.

Slaughter, S., & Rhoades, G. (2004). *Academic capitalism and the new economy: Markets, state, and higher education*. Johns Hopkins University Press.

Superintendencia de Educación Superior. (2021). *Resolución Exenta 510: Normativa contable para las instituciones de educación superior no estatales*.

Superintendencia de Educación Superior. (2025). *Salud financiera: Gastos y costos en la educación superior chilena, 2012–2024. Estudio 2025*.

Superintendencia de Educación Superior. (n.d.). *Norma de Carácter General N. 1: Establece normas sobre obligación de informar de las instituciones de educación superior*.

Superintendencia de Educación Superior & Contraloría General de la República. (2022). *Plan de cuentas y diccionario de cuentas para las universidades del Estado* (Oficio 292).

Valenzuela Nieto, I. (2015). *La prohibición a los fines de lucro y propuestas de gobierno para las universidades chilenas* (Propuestas de Política Pública No. 9). Centro de Estudios Públicos.

Webber, D. A., & Ehrenberg, R. G. (2010). Do expenditures other than instructional expenditures affect graduation and persistence rates in American higher education? *Economics of Education Review*, *29*(6), 947–958.

Williamson, C., Muñoz, I., & Villalobos, M. (2024). *Gratuidad y financiamiento universitario* (Documento de Trabajo No. 139). Clapes UC.